

NOTTINGHAM WATNALL, United Kingdom

WMO: 033540

Lat: 53.0056N

Lon: 1.2511W

Elev: 384

StdP: 14.49

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	28.8	21.0	16.0	30.7	23.5	18.0	31.7	24.7	46.1	22.8	46.4	6.5	40	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	79.3	65.2	75.4	63.1	72.0	61.7	66.7	75.4	64.7	72.5	63.1	69.9	8.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	89.4	69.8	61.8	83.7	67.6	60.3	79.2	65.9	31.5	75.8	29.9	72.6	28.7	70.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.3	16.3	DB	22.5	85.6	3.4	3.7	20.1	88.3	18.1	90.4	16.1	92.5	13.6	95.2	(4)
(5)			WB	21.7	70.8	3.4	2.5	19.2	72.6	17.2	74.1	15.3	75.5	12.8	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	39.7	40.5	43.4	47.6	52.9	58.2	62.2	61.7	57.1	51.3	44.5	40.5	(6)	
	DBStd	9.57	5.37	5.62	5.33	5.33	5.34	4.82	4.79	4.31	4.21	4.93	5.31	6.13	(7)	
	HDD50	1457	320	269	214	107	29	1	0	0	1	42	175	299	(8)	
	HDD65	5531	784	687	671	522	376	212	117	122	240	425	614	761	(9)	
	CDD50	1463	1	2	8	35	119	247	377	364	214	82	11	3	(10)	
	CDD65	62	0	0	0	0	1	7	29	21	3	1	0	0	(11)	
	CDH74	469	0	0	0	1	11	74	228	132	21	2	0	0	(12)	
	CDH80	78	0	0	0	0	0	12	40	25	1	0	0	0	(13)	
(14) Wind	WSAvg	8.0	9.1	9.3	9.0	8.2	7.8	7.2	6.8	6.7	7.0	8.0	8.1	8.6	(14)	
(15) Precipitation	PrecAvg	28.1	2.6	2.0	2.0	2.0	2.1	2.2	2.2	2.4	2.4	2.4	2.6	2.8	(15)	
	PrecMax	36.9	5.0	7.8	4.5	4.5	5.3	6.5	5.8	5.0	5.6	5.7	5.2	7.1	(16)	
	PrecMin	20.5	0.6	0.4	0.2	0.3	0.4	0.2	0.4	0.2	0.4	0.4	1.0	0.7	(17)	
	PrecStd	4.2	1.0	1.4	1.1	1.1	1.2	1.5	1.3	1.1	1.5	1.3	1.0	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	56.5	61.0	70.2	75.4	81.4	84.9	83.9	77.0	67.5	59.3	56.1	(19)	
		MCWB	51.6	51.7	51.9	56.7	61.8	65.8	67.1	66.6	63.7	60.9	56.2	52.9	(20)	
	2%	DB	51.9	52.9	56.9	63.7	70.8	75.3	79.7	78.0	71.1	63.1	56.6	53.9	(21)	
		MCWB	49.1	49.2	50.4	53.9	59.6	62.5	65.2	65.3	62.0	58.1	54.0	51.5	(22)	
	5%	DB	49.9	50.8	54.1	59.9	66.5	71.1	76.0	73.1	67.5	60.7	54.6	51.8	(23)	
		MCWB	47.4	47.6	48.8	51.7	57.0	60.5	63.6	62.9	59.8	56.6	52.0	49.5	(24)	
	10%	DB	47.9	48.8	51.8	56.8	62.8	67.8	72.0	70.1	64.7	58.7	52.4	49.4	(25)	
		MCWB	45.7	45.8	47.3	50.2	54.8	59.0	61.8	61.4	58.2	55.3	50.3	47.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.4	53.6	58.4	63.8	68.2	69.8	69.5	65.8	62.0	57.1	53.6	(27)	
		MCDB	53.9	55.0	58.1	67.6	72.3	78.7	81.4	77.8	73.0	66.3	58.6	55.2	(28)	
	2%	WB	49.7	49.9	51.6	55.3	60.6	64.0	66.6	66.5	63.3	59.5	54.6	51.9	(29)	
		MCDB	51.5	52.5	55.3	61.7	69.2	72.3	76.3	73.6	69.0	62.0	56.2	53.7	(30)	
	5%	WB	47.9	48.1	49.7	53.0	58.2	61.9	64.7	64.4	61.2	57.6	52.5	49.9	(31)	
		MCDB	49.5	50.6	53.4	58.8	65.0	68.9	72.8	71.7	65.7	59.9	54.0	51.7	(32)	
	10%	WB	45.9	46.2	48.0	50.9	55.9	60.0	63.0	62.7	59.5	55.8	50.6	47.6	(33)	
		MCDB	47.8	48.4	51.3	56.0	61.4	66.3	70.2	68.6	63.3	58.2	52.1	49.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.6	11.9	14.2	15.0	14.9	15.6	14.9	13.6	11.2	9.1	8.1	(35)	
		MCDBR	10.5	11.5	15.1	18.8	20.9	21.1	22.1	19.9	17.6	12.6	10.2	10.3	(36)	
	5% WB	MCWBR	9.7	8.9	10.0	10.8	11.7	11.0	9.8	9.1	9.6	8.5	8.9	9.3	(37)	
		MCWBR	10.3	10.8	13.8	17.7	19.2	18.9	19.7	17.3	15.7	11.9	10.0	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.321	0.336	0.368	0.395	0.398	0.399	0.399	0.386	0.367	0.348	0.323	0.309	(40)	
		taud	2.349	2.386	2.295	2.232	2.263	2.284	2.288	2.349	2.406	2.437	2.405	2.328	(41)	
	Ebn at Noon	Ebn at Noon	207	244	258	265	270	270	268	265	256	236	207	188	(42)	
		Edh at Noon	19	24	33	39	40	40	39	35	29	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	225	443	783	1205	1502	1578	1529	1260	936	540	278	182	(44)	
	RadStd	RadStd	21	52	94	139	98	152	163	86	85	46	31	19	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page